PHISHING EMAIL DETECTION & AWARENESS REPORT

Future Interns – Cyber Security Task 2

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1. Executive Summary

Phishing remains one of the most common and successful cyber threats targeting organizations worldwide. Unlike technical attacks that exploit system vulnerabilities, phishing attacks manipulate human psychology using urgency, fear, and deception.

This report analyzes selected phishing email samples to identify common attack indicators, classify their risk level, and provide clear awareness guidance for employees. The objective is to help users recognize fraudulent emails and reduce the risk of credential theft, financial loss, and data compromise.

2. Scope & Objective

This assessment focuses on analyzing phishing email samples to identify common red flags and attack patterns. The objective is to classify email risk levels and create a practical awareness guide that organizations can use for employee training.

Scope Includes:

- Analysis of phishing email content
- Identification of suspicious sender domains
- Inspection of embedded links
- Classification of risk level
- Creation of prevention guidelines

Scope Excludes:

- Email server penetration testing
- Exploitation of links or attachments
- Malware analysis
- Any intrusive or illegal activity

This task is focused on security awareness and risk identification.

3. Phishing Email Sample 1 – Account Verification Scam

Email Subject:

⚠️ Urgent: Your Account Will Be Locked

Email Body Summary:

The email claims suspicious activity was detected on the user's account and urges immediate verification using a provided link.

Suspicious Link:

http://secure-account-verify[.]com

Identified Phishing Indicators:

- Generic greeting ("Dear User")
- Urgency and fear-based language
- Threat of account suspension
- Suspicious domain name
- No official company branding

How the Attack Works:

The attacker creates a sense of panic by claiming suspicious activity. The victim is pressured to click a malicious link, which likely leads to a fake login page designed to steal credentials.

Risk Classification:

High Risk – Phishing

4. Phishing Email Sample 2 – Invoice Payment Scam

Email Subject:

Invoice #84321 – Immediate Payment Required

Sender:

billing@paypa1-support.com

Email Body Summary:

The email claims a recent transaction failed and instructs the user to download an invoice or complete payment immediately.

Suspicious Link:

http://paypal-secure-billing[.]com

Identified Phishing Indicators:

- Spoofed domain name ("paypa1" instead of "paypal")
- Financial urgency and pressure
- Unexpected invoice request
- Suspicious link not matching official PayPal domain
- Threat of account restriction

How the Attack Works:

The attacker impersonates a trusted financial brand and uses urgency related to payments. Victims may panic and either download a malicious attachment or enter sensitive financial information on a fake website.

Risk Classification:

High Risk – Phishing

5. Email Header & Domain Analysis

Although full raw email headers were not available for forensic extraction, phishing emails of this nature typically reveal inconsistencies when analyzed using email header analysis tools.

Common header anomalies observed in phishing emails include:

- Mismatch between the “From” address and the actual sending domain
- Suspicious or unknown sending IP addresses
- SPF (Sender Policy Framework) failure
- DKIM authentication failure
- Reply-To address different from the sender address

In the second sample email, the sender domain “paypa1-support.com” imitates a legitimate brand by replacing the letter “l” with the number “1”. This technique is known as domain spoofing or typosquatting.

Such domain manipulation is a strong indicator of phishing intent and is commonly used to deceive users into trusting fraudulent communication.

Domain inspection of the embedded links further indicates that the URLs do not belong to official corporate domains and are likely malicious landing pages designed for credential harvesting or financial fraud.

6. Common Phishing Indicators Identified

Based on the analysis of both samples, the following phishing indicators were identified:

- Generic greetings (e.g., “Dear User”)
- Urgency and fear-based messaging
- Threats of account suspension or restriction
- Suspicious or misspelled domain names
- Financial pressure tactics
- Embedded links leading to unknown domains
- Lack of official branding or verification details

These characteristics are consistent with real-world phishing campaigns targeting corporate and individual users.

7. Risk Classification Model

The following model was used to classify email risk:

Safe – Verified sender, legitimate domain, no suspicious indicators
Suspicious – Minor inconsistencies requiring manual verification
Phishing – Clear malicious intent with multiple deception indicators

Both analyzed email samples fall under the "Phishing – High Risk" category due to multiple confirmed red flags.

8. Prevention & Awareness Guidelines

Organizations can significantly reduce phishing risk through user awareness and procedural controls.

Recommended practices include:

- Always verify the sender's domain carefully
- Hover over links before clicking to inspect the full URL
- Never share passwords, OTPs, or confidential data via email
- Confirm financial or urgent requests through official channels
- Report suspicious emails to the IT or security team immediately
- Enable Multi-Factor Authentication (MFA) for sensitive accounts

User education remains one of the strongest defenses against phishing attacks.

9. Employee Do's and Don'ts

Do:

- Verify unexpected requests before responding
- Double-check domain spellings
- Report suspicious emails promptly
- Use strong passwords and MFA

Don't:

- Click unknown or suspicious links
- Download unexpected attachments

- Share confidential information via email
- Respond to fear-based or urgent pressure tactics

10. Conclusion

Phishing attacks rely primarily on psychological manipulation rather than technical exploitation. By creating urgency, impersonating trusted brands, and embedding deceptive links, attackers successfully trick users into compromising sensitive information.

The analysis conducted in this report demonstrates how easily phishing indicators can be identified when proper awareness techniques are applied.

Continuous employee training, email verification procedures, and strong reporting mechanisms are essential to reducing phishing-related incidents within organizations.

Proactive awareness remains the most effective defense against phishing threats.